Connectome Brain Age: Audit, Revised Pipeline and Results

Participant-nested evaluation of DWI, signed fMRI and early fusion

Audit findings → training-only transformations → nested prediction

Matched Ridge and full-budget GINE → calibration and remaining validation

Current development results first; historical benchmarks and reconstructed attribution in the appendix.

Three points to take away

- The audit identified full-sample preprocessing, outer-score threshold selection and mixed calibration provenance.
- The revised evaluation nests preprocessing, selection and stopping within participant-level training partitions.
- Matched Ridge and completed GINE arms are available; clinical source validation and anatomical interpretation remain unresolved.

Implementation validation supports the evaluation design; it does not verify the underlying clinical data.

audit_findings.tsv; audit_to_implementation.csv; completed-run audit.json

Audit findings and the implemented response

Finding	Revised behavior
Global ComBat + full-sample feature mask	Training-only QC, ComBat and scaling at every fit boundary
Threshold chosen using reported OOF scores	Choose sparsity within inner validation
Stopping and selection need separation	Separate inner stopping split; fixed-epoch outer refit
CN final predictions include in-sample fits	Evaluate held-out predictions; calibrate from training inner OOF
Unverified pairing and reconstructed attribution	Explicit pairing exclusions; original checkpoint attribution

reports/pipeline_audit/audit_findings.tsv; reports/new_pipeline/audit_to_implementation.csv

The revised pipeline: fitting stays within training

1 Register

Raw connectomes

Metadata evidence

Canonical participant groups

→

2 Split

Frozen outer folds

CN training reference

Separate inner / stopping folds

→

3 Fit

Training-only QC

Additive ComBat

Scaling + selected model

→

4 Evaluate

Outer held-out age

Training-OOF calibration

Saved states and audits

No outer-test age enters feature transformation, configuration choice or early stopping.

code/brainage_pipeline/{data,splits,preprocessing,engine,calibration}.py

Current eligible populations and matched comparison

Population	Scans	Participants	CN scans
DWI	919	670	723
fMRI	575	447	441
paired	318	252	237

Matched CN performance uses 237 scans / 189 participants; the full paired registry has 318 scans / 252 participants.

reports/new_pipeline/sample_counts.csv

Cross-modality pairing: conservative exclusions

575 keys

Common scan keys

Unique source matching

Metadata agreement required

→

250 excluded

Age disagreement

Tolerance: 0.01 years

No age averaging

→

7 excluded

Insufficient paired CN batch support

Nested fits must be feasible

→

318 retained

252 participants

237 CN scans

Dates still require verification

Agreeing keys and metadata do not establish contemporaneous DWI and fMRI acquisition.

reports/new_pipeline/pairing_counts.csv; code/brainage_pipeline/data.py

Inputs, transforms and notation

Symbol / object	Definition
A^D, A^f	Symmetric 84 × 84 DWI and fMRI connectivity matrices
$x^D \in R^{3486}$	Upper-triangle DWI edges; fixed log1p transformation
$x^f \in R^{3486}$	Upper-triangle signed fMRI edges; clipped Fisher-z transformation
b, y, g	Known acquisition batch; age target; canonical participant group
CN reference	Diagnosis defines reference eligibility; it is not an imaging predictor

Age is the supervised target; no sex, APOE or metabolic variables enter the age-prediction features.

code/brainage_pipeline/data.py; code/final/{harmonization,foldwise_fmri}.py

Feature QC and edge reconstruction are training-only

- Apply fixed modality transforms before learning feature statistics.
- Fit feature masks on the corresponding training rows, including nonconstant and residual-variance checks.
- Fit harmonization and StandardScaler only after training-only edge selection.
- Restore graph values to their original ROI-pair positions; excluded edges map to scaled zero and cannot enter topology.
- Frozen transforms reject unsupported batches and invalid input instead of silently adapting to test data.

code/final/harmonization.py; code/brainage_pipeline/preprocessing.py

Harmonization: compare declared modes

- No-ComBat control retains the same training-only QC and scaling boundaries.
- Training-age mean-only ComBat estimates additive batch offsets using training ages.
- Held-out transformation requires imaging and known batch metadata, without chronological age.
- Batch-only ComBat is an available sensitivity mode; the completed matched comparisons report none versus age_mean.
- Known-batch eligibility restricts the population; this does not evaluate generalization to an unseen acquisition batch.

Reduced technical predictability is not proof that all biological variation is preserved.

code/brainage_pipeline/preprocessing.py; code/final/harmonization.py

Additive harmonization: fit once, apply frozen offsets

Step	Operation
Fit on training rows	Estimate training-age-adjusted $\hat{\gamma}[b,j]$ and pooled scale $\hat{\sigma}[j]$
Mean-only adjustment	$x^{H}[i,j] = x[i,j] - \hat{\sigma}[j]\,\hat{\gamma}[b_i,j]$
Training scaling	$z[i,j] = (x^{H}[i,j] - \hat{\mu}_{train}[j]) / \hat{s}_{train}[j]$
Apply to held-out rows	Reuse mask, offsets and scaler; do not re-estimate parameters
Age boundary	Training age informs offsets; test age is unnecessary for this transform

The mean-only age terms cancel during application; this formula is not the general location-scale ComBat rule.

code/final/harmonization.py::_adjust; code/brainage_pipeline/preprocessing.py

Harmonization diagnostics require held-out evidence

- Compare age-prediction performance under no-ComBat and additive harmonization.
- Assess technical and sex predictability on held-out representations with training-fitted probes.
- Fit PCA to training representations and project held-out scans in a shared coordinate system.
- Inspect cohort-specific age distributions and within-cohort cBAG residual trends.
- Technical batch, cohort and biology overlap; visual mixing and probe scores alone cannot establish preservation.

code/brainage_pipeline/diagnostics.py; code/brainage_pipeline/reporting.py

Nested evaluation: outer test is reserved for scoring

Outer folds

5 participant folds

All visits grouped

Train on CN reference

→

Inner folds

3 participant folds

Fit preprocessing anew

Select α or graph sparsity

→

Outer refit

Refit training transforms

Fit selected configuration

Use fixed GINE epochs

→

Outer test

One held-out prediction

Frozen calibration

No selection or stopping

DWI, fMRI and fusion share paired observations and frozen participant folds.

code/brainage_pipeline/{splits,engine}.py; configs/ridge.json

GINE stopping is separate from inner validation

Early training	→	Stopping split	→	Inner refit	→	Outer refit
Subset of inner training Fit own preprocessing Optimize model weights		Participant-disjoint Select epoch count No inner validation access		Refit all inner training Use selected epochs Then score inner validation		All outer CN training Median selected epochs No outer early stopping

Maximum 250 epochs, patience 25; stored histories distinguish optimization, stopping and refit stages.

code/brainage_pipeline/engine.py; code/brainage_pipeline/gnn.py

Ridge: a fully nested imaging baseline

- Fit a linear age predictor using transformed, training-standardized imaging edges.
- Choose α from 0.1, 1, 10, 100, 1,000, 10,000 and 100,000 using inner OOF MAE.
- Re-estimate preprocessing and Ridge on all outer CN training rows after selecting α.
- Early fusion concatenates independently standardized DWI and fMRI features.
- Evaluate raw held-out age predictions; BAG calibration is a separate downstream operation.

$\hat{\beta} = \arg\min_{\beta} \|y - X\beta - c\|^2 + \alpha\|\beta\|^2$; α is never selected from outer-test scores.

Graph construction: signed features and per-scan topology

Component	Current definition
Nodes	84 ROIs; each node contains its complete standardized connectivity profile
DWI topology	Positive raw connectivity; retain per-scan top 25% or 50% of eligible edges
fMRI topology	Rank raw magnitude; retain positive and negative connections
Edge attributes	Signed transformed / standardized edge values
QC and sparsity	Training mask removes invalid edges; sparsity fraction selected by inner validation
Directionality	Each selected undirected pair becomes two directed edges for message passing

Unimodal node matrix: 84 × 84; edge attributes: one channel.

Current GINE architecture

Graph input	GINE layers	Readout	Age head
Node profiles	MLP message updates	Global mean pooling	Width → width/2 → 1
Signed edge attributes →	Batch normalization →	Optional residual skips →	Training target scaling
Selected topology	ReLU + dropout	No max-pooling branch	Return age in years

AdamW: learning rate 5×10^{-4}; weight decay 5×10^{-4}; dropout 0.3; batch size 32.

code/brainage_pipeline/gnn.py::GINE

GINE notation and capacity arms

Quantity	Definition
Message update	$h_v' = \mathrm{MLP}((1 + \varepsilon)h_v + \Sigma_u \mathrm{ReLU}(h_u + E(e_{uv})))$
Layer output	$\mathrm{ReLU}(\mathrm{BN}(h_v') + S(h_v))$ for residual arms; omit S otherwise
Readout / age	$g = \mathrm{mean}_v\, h_v$; age = μtrain + σtrain × head(g)
Prespecified capacities	1 layer × 16; 2 × 32; 3 × 64; 2 × 32 with residual skips
Within-arm selection	Capacity fixed for each ablation; select sparsity inside inner validation

Outer-score capacity comparisons are development analyses; choosing a winner needs further validation.

code/brainage_pipeline/gnn.py; configs/gine_*.json

Fusion now combines modalities at the input level

DWI branch	fMRI branch	Early fusion	One predictor
Training-fit transform Standardized profile Modality-specific mask	→ Training-fit transform Signed profile Modality-specific mask	→ Ridge: concatenate edges GINE: concatenate profiles Union graph topology	→ One Ridge model or one GINE encoder Matched participant folds

Historical weighted / late Ridge and dual-encoder GNN results are retained only in the appendix.

code/brainage_pipeline/{preprocessing,gnn,engine}.py

Fusion dimensions and modality alignment

Object	Current representation
Ridge fusion vector	Concatenated retained DWI and fMRI features; width depends on training masks
GINE node input	[DWI profile | fMRI profile] $\in R^{168}$ for each of 84 nodes
GINE edge attributes	[standardized DWI value, standardized fMRI value] $\in R^{2}$
Fusion topology	Union of per-modality selected ROI pairs; bidirectional message passing
Comparison boundary	Same observations and participant folds as the unimodal matched models

code/brainage_pipeline/gnn.py::graphs; preprocessing.py::matrix

BAG calibration uses the outer training reference

Inner OOF

CN outer-training rows

Selected model candidate

Predict without own inner fit

→

Fit age trend

BAG = prediction − age

Global and cohort trends

Record sparse-cohort fallback

→

Outer prediction

Model fitted on outer train

Predict held-out scans

Keep age-model output raw

→

Apply correction

Subtract frozen age trend

Save global and cohort cBAG

No test-coefficient fitting

Calibration uses selected-candidate inner OOF predictions; external confirmation is still needed.

code/brainage_pipeline/calibration.py; engine.py

BAG and cBAG: equations and reference limits

Quantity	Definition
Raw BAG	$BAG_i = \hat{y}_i - y_i$
Training-reference trend	Fit BAGinner-OOF = a + b × age on CN outer-training rows
Global cBAG	$cBAGglobal,_i = \hat{y}_i - y_i - (aglobal + bglobal \times y_i)$
Cohort-aware cBAG	Use cohort-specific a,b when ≥10 training reference scans and age varies
Fallback / age usage	Sparse cohorts use global trend; test age is used for BAG arithmetic, not imaging transformation

Neither low pooled age correlation nor corrected BAG alone establishes a clinical biomarker.

code/brainage_pipeline/calibration.py::BiasCorrection

Prediction and calibration provenance is auditable

- Save original outer model weights, preprocessing state and selected configuration.
- Persist training indices, participant folds, inner OOF calibration predictions and coefficients locally.
- Audit split boundaries, training-only fit indices, complete outer predictions and reconstructed cBAG arithmetic.
- Record data/source fingerprints; incompatible resume requests are rejected.
- Synthetic boundary tests validate the software; source diagnoses, dates, protocols and ROI ordering need independent evidence.

code/brainage_pipeline/audit.py; code/test_pipeline.py; completed-run audit.json

Matched Ridge: current held-out age performance

Modality	Additive MAE	95% CI	R^2	No-ComBat MAE
DWI	6.22	5.59–6.90	0.662	6.29
fMRI	7.95	7.15–8.77	0.466	7.89
fusion	6.41	5.79–7.04	0.658	6.41

237 CN scans / 189 participants; fusion does not improve on DWI Ridge in this matched population.

reports/new_pipeline/matched_ridge/performance.csv

Full-budget comparison: Additive harmonization

Model arm	DWI MAE	fMRI MAE	Fusion MAE
Ridge	6.22	7.95	6.41
GINE 1x16	7.83	9.39	8.11
GINE 2x32	7.84	9.85	7.83
GINE 3x64	8.55	9.80	8.30
GINE 2x32_skip	8.12	9.66	7.92

All arms use the same 237 CN scans / 189 participants and participant folds; MAE in years.

reports/new_pipeline/gine_study/ridge_gine_comparison.csv

Full-budget comparison: No ComBat

Model arm	DWI MAE	fMRI MAE	Fusion MAE
Ridge	6.29	7.89	6.41
GINE 1x16	7.48	9.46	8.05
GINE 2x32	7.65	9.99	7.86
GINE 3x64	7.94	9.85	8.17
GINE 2x32_skip	7.62	10.09	7.45

All arms use the same 237 CN scans / 189 participants and participant folds; MAE in years.

reports/new_pipeline/gine_study/ridge_gine_comparison.csv

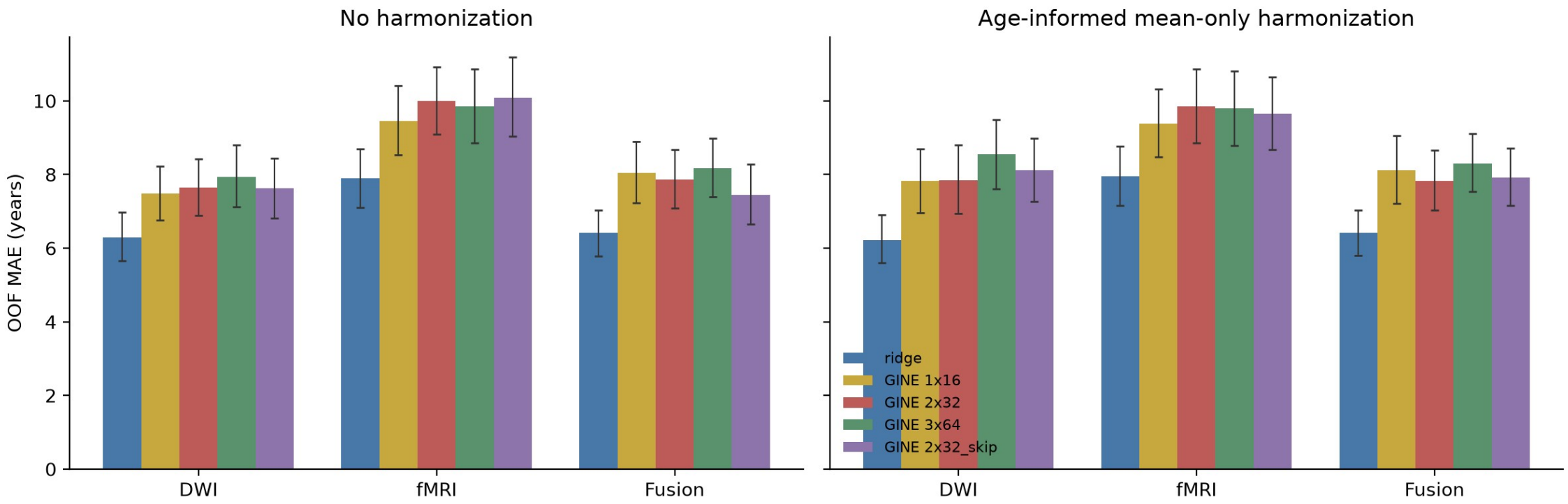
GINE 2×32 versus Ridge: paired differences

Modality	ΔMAE (GINE − Ridge)	95% fixed-OOF CI
DWI	1.62	0.81–2.44
fMRI	1.90	1.02–2.72
fusion	1.42	0.69–2.17

Positive values favor Ridge. These intervals condition on trained OOF models; they do not include model refitting.

reports/new_pipeline/gine_study/paired_MAE_differences.csv

Capacity and residual comparisons



Report every prespecified arm; capacity selection based on these outer scores requires further validation.

reports/new_pipeline/gine_study/architecture_comparison.png; performance.csv

Additive ComBat versus no ComBat: paired sensitivity

Arm	Modality	ΔMAE (additive − none)	95% CI
ridge	DWI	-0.065	-0.167–0.031
ridge	fMRI	0.059	-0.007–0.124
ridge	fusion	0.004	-0.109–0.118
gine_2x32	DWI	0.193	-0.482–0.884
gine_2x32	fMRI	-0.141	-0.904–0.615
gine_2x32	fusion	-0.032	-0.632–0.516

No consistent accuracy gain across these contrasts; the preprocessing mode is declared rather than selected on outer scores.

reports/new_pipeline/gine_study/harmonization_paired_MAE_differences.csv

Cohort-specific performance: DWI Ridge

Cohort	CN scans	Participants	MAE (yr)	R^2
ADNI	125	77	5.08	0.161
ADRC	106	106	7.59	0.333
HABS	6	6	5.91	0.313

Pooled performance mixes cohort age distributions; HABS has only six retained CN scans.

reports/new_pipeline/matched_ridge/performance.csv

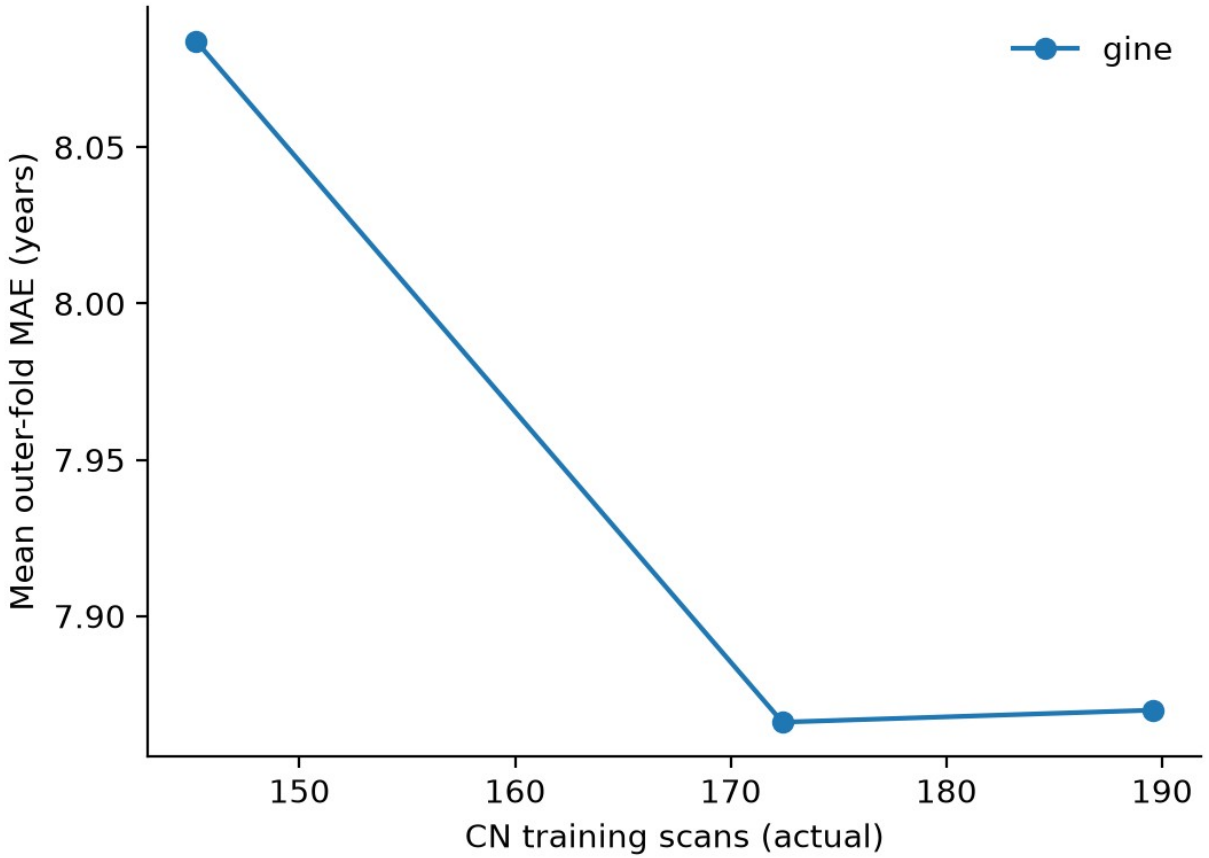
Calibration diagnostics: inspect each cohort

DWI Ridge	r(age, BAG)	r(age, global cBAG)	r(age, cohort cBAG)
Pooled	-0.613	0.027	0.022
ADNI	-0.683	-0.339	0.024
ADRC	-0.814	-0.530	0.101
HABS	-0.927	-0.762	-0.762

These are held-out residual diagnostics, not evidence of clinical validity or complete age independence.

reports/new_pipeline/matched_ridge/age_calibration.csv

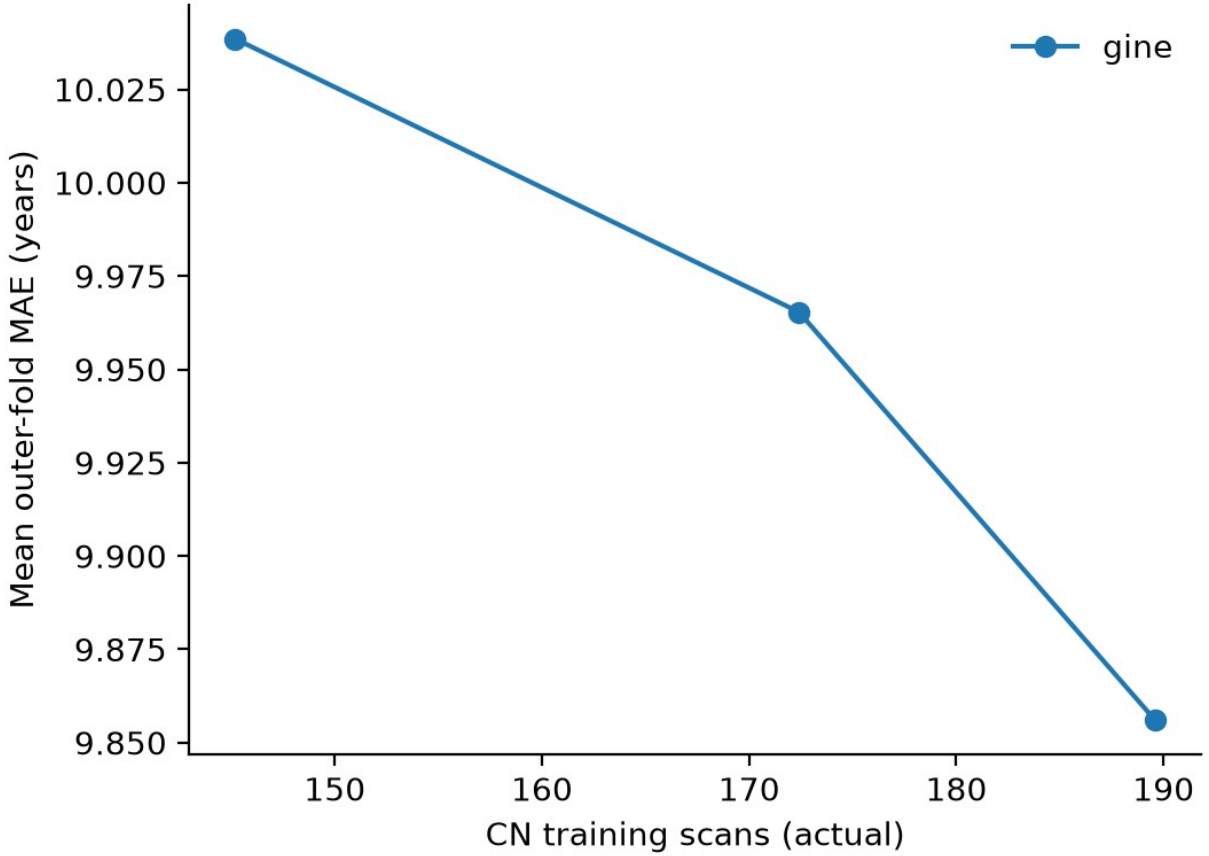
Training size: DWI GINE 2×32



Mean outer-fold MAE, not pooled MAE; fixed outer tests. 25%, 50% and 60% failed batch-support preflight.

gine_study/training_size_gine_2x32/training_size_metrics.csv | additive mode

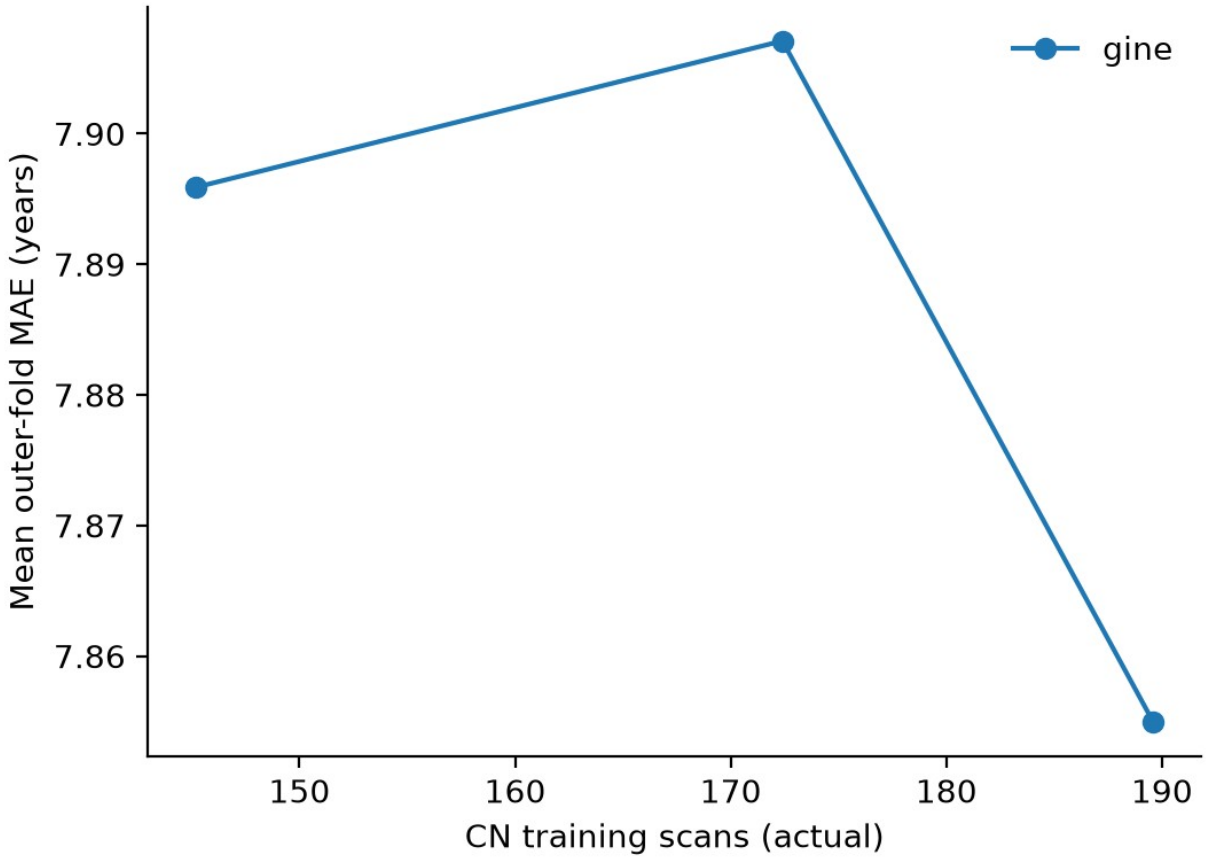
Training size: fMRI GINE 2×32



Mean outer-fold MAE, not pooled MAE; fixed outer tests. 25%, 50% and 60% failed batch-support preflight.

gine_study/training_size_gine_2x32/training_size_metrics.csv | additive mode

Training size: fusion GINE 2×32



Mean outer-fold MAE, not pooled MAE; fixed outer tests. 25%, 50% and 60% failed batch-support preflight.

gine_study/training_size_gine_2x32/training_size_metrics.csv | additive mode

Training-size comparison: Ridge and GINE 2×32

Modality	Fraction	Mean CN train scans	Ridge MAE	GINE MAE
DWI	75%	145.2	6.41	8.08
DWI	90%	172.4	6.24	7.87
DWI	100%	189.6	6.23	7.87
fMRI	75%	145.2	7.93	10.04
fMRI	90%	172.4	8.07	9.97
fMRI	100%	189.6	7.96	9.86
fusion	75%	145.2	6.51	7.90
fusion	90%	172.4	6.46	7.91
fusion	100%	189.6	6.41	7.85

Additive mode; arithmetic mean of five outer-fold MAEs. Fractions refer to participant groups, not scan counts.

gine_study/training_size_{ridge,gine_2x32}/training_size_metrics.csv

Learning diagnostics: DWI additive-mode GINE

Arm	Parameters	Median epoch	Train MAE	Stopping MAE
1x16	1977	74	4.84	8.32
2x32	6793	30	4.94	8.70
3x64	29161	18	4.06	8.17
2x32_skip	10505	21	5.16	8.58

Training and stopping MAEs are inner early-fit diagnostics, not outer-test performance.

reports/new_pipeline/gine_study/early_stopping_summary.csv

What has completed, and what remains unverified

- Matched Ridge: 30 outer tasks; complete predictions for six modality × preprocessing combinations.
- Four full-training GINE capacity / residual arms: each completed 30 outer tasks and passed its artifact audit.
- 75% and 90% training-size runs completed for Ridge and GINE 2×32; 25%, 50% and 60% failed batch-support preflight.
- Short-budget GINE validation is software evidence and is excluded from trained-model comparisons.
- Clinical labels, acquisition dates/protocols and ROI ordering remain unverified.

Status is taken from per-run completion records, rather than older summary text.

gine_study/study_status.csv; matched_ridge and GINE run audit.json

Interpretability uses original saved outer checkpoints

Load		Verify		Compute		Interpret
Original outer weights Exact preprocessing Matching selected config	→	Held-out sample identity Prediction consistency Source / state provenance	→	Gradient × input Node / edge sensitivity Held-out embeddings	→	Numeric ROI indices first Labels require verified order No causal claims

The current method is local gradient-times-input sensitivity; archived integrated-gradient figures used reconstructed models.

code/brainage_pipeline/downstream.py

Embeddings and anatomical labels need separate checks

- Extract embeddings from the original outer checkpoint for training and held-out observations.
- Fit the PCA basis to training embeddings separately for each fold, then project that fold's held-out observations.
- Do not pool raw embedding coordinates across independently fitted models.
- Age, cohort, sex and verified clinical variables are overlays rather than age-prediction inputs.
- Keep ROI labels numeric until the 84-region order is verified; network assignments are not guessed.

code/brainage_pipeline/downstream.py; diagnostics.py

Clinical analysis: implemented interface, gated inference

- Verify source diagnosis codes and the CN reference before interpreting clinical group effects.
- Confirm imaging-to-clinical visit alignment, APOE coding, diabetes definition and HbA1c units.
- The interface supports complete-case analyses with cohort, age and sex covariates, participant-cluster uncertainty and BH adjustment.
- Clinical inference requires a reviewed specification and nonpilot held-out predictions.
- Generic diabetes fields do not establish type 2 diabetes; missing definitions are not filled in by assumption.

No verified clinical association or anatomical finding is claimed from these implementation checks.

code/brainage_pipeline/clinical.py; configs/clinical_spec.json

Limits that still determine interpretation

- This is a development reanalysis following inspection of earlier results, rather than untouched external validation.
- Known-batch eligibility and pairing exclusions produce a restricted matched population.
- Canonical IDs, diagnosis provenance, visit dates and ADRC acquisition protocols need source confirmation.
- Bootstrap intervals describe uncertainty in fixed OOF predictions and omit refitting uncertainty.
- Residual cohort effects and unverified ROI order constrain biomarker and anatomical conclusions.

code/brainage_pipeline/README.md; gine_study/README.md

Next decisions for the project

- Prioritize source verification of reference diagnosis, participant identity and imaging / clinical visit dates.
- Resolve metadata conflicts with source evidence before expanding the paired population.
- Use a prespecified validation design before choosing a final capacity or preprocessing mode from development results.
- Verify ROI ordering before attaching anatomical names to checkpoint-based sensitivity.
- Run clinical association analyses only after definitions, units and alignment have been reviewed.

Reproduction and result provenance

Artifact	Authoritative location
Current entry point	code/run_pipeline.py
Methods / configuration	code/brainage_pipeline/ and configs/
Matched Ridge aggregates	reports/new_pipeline/matched_ridge/
Full-budget study aggregates	reports/new_pipeline/gine_study/
Original audit evidence	reports/pipeline_audit/audit_findings.tsv
This editable deck builder	code/final/slides/update_comprehensive_pipeline.py

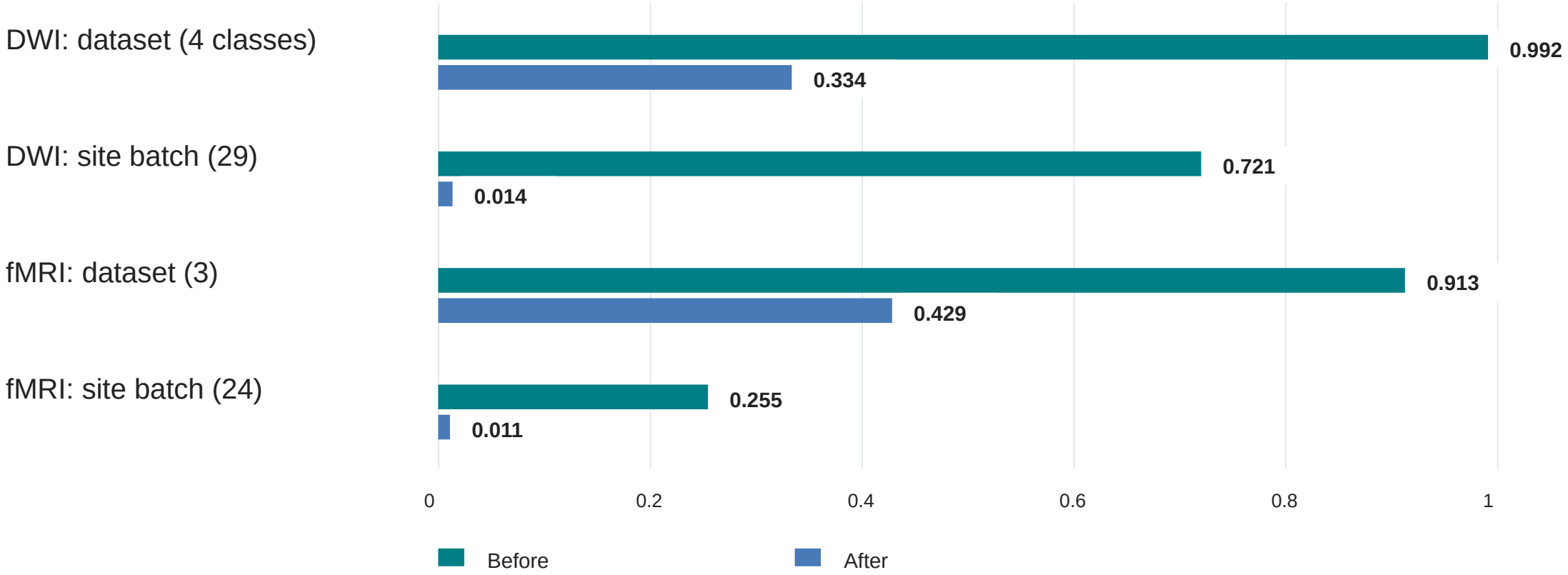
Slide tables are loaded from aggregate CSVs; historical figures below retain their original numbers.

Local runtime predictions/checkpoints stay under ignored results/final/new_pipeline/

Appendix: historical benchmarks and reconstructed attribution

- The following figures preserve the earlier full-sample harmonization and 387-CN-scan benchmark.
- They use different populations and evaluation procedures from the current 237-CN-scan matched results.
- Historical graph thresholds were compared using their reported OOF scores.
- Historical attribution used reconstructed models; it is not current original-checkpoint sensitivity.
- These archived numbers are retained for context and must not be read as updated validation results.

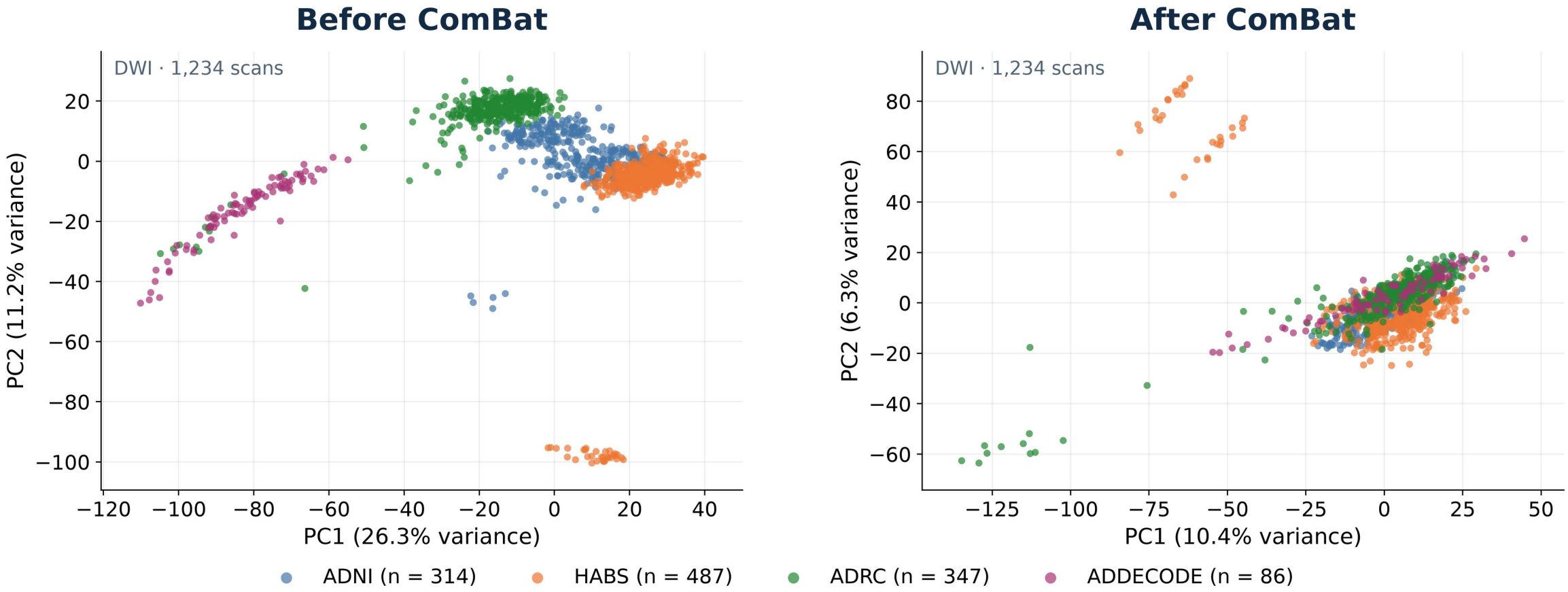
Historical: Dataset and site predictability decrease sharply



Balanced accuracy; lower technical predictability supports reduced batch information.

report_tables/Table3_harmonization.csv

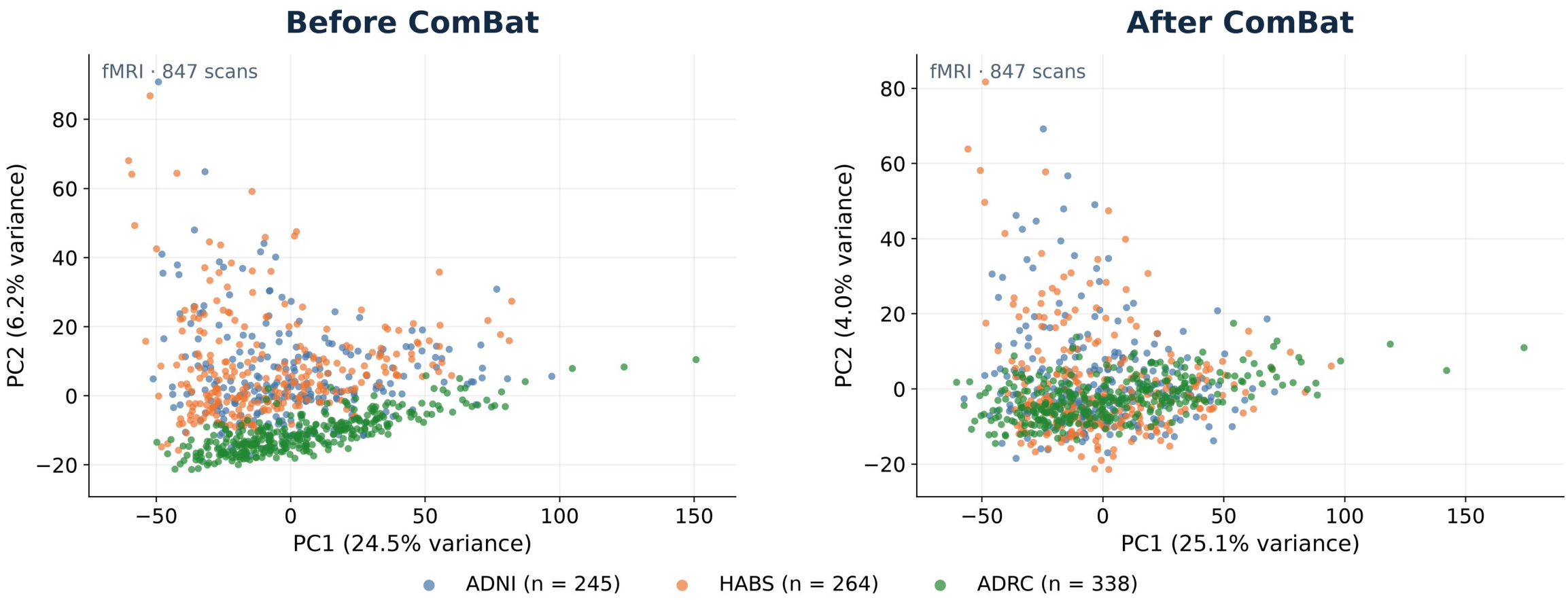
Historical: PCA by dataset: DWI



Separate standardized PCA fits; axes are specific to each panel. All eligible scans are shown.

results/checks/PCA/figures/DWI_PCA_by_dataset_before_after.png

Historical: PCA by dataset: fMRI



Separate standardized PCA fits; axes are specific to each panel. All eligible scans are shown.

Final fMRI run: 20261028T234533Z | results/final/fMRI/harmonization_evaluation/

Historical: Biological signal preservation is mixed

Diagnostic	DWI before → after	fMRI before → after
Sex balanced accuracy	0.782 → 0.766	0.631 → 0.640
Diagnosis balanced accuracy	0.294 → 0.151	0.300 → 0.227
Age MAE (years)	6.22 → 6.71	7.64 → 8.00
Age R^2	0.588 → 0.516	0.379 → 0.307

Sex prediction is broadly retained. Age prediction worsens and diagnosis prediction decreases; these diagnostics do not establish complete biological preservation.

report_tables/Table3_harmonization.csv

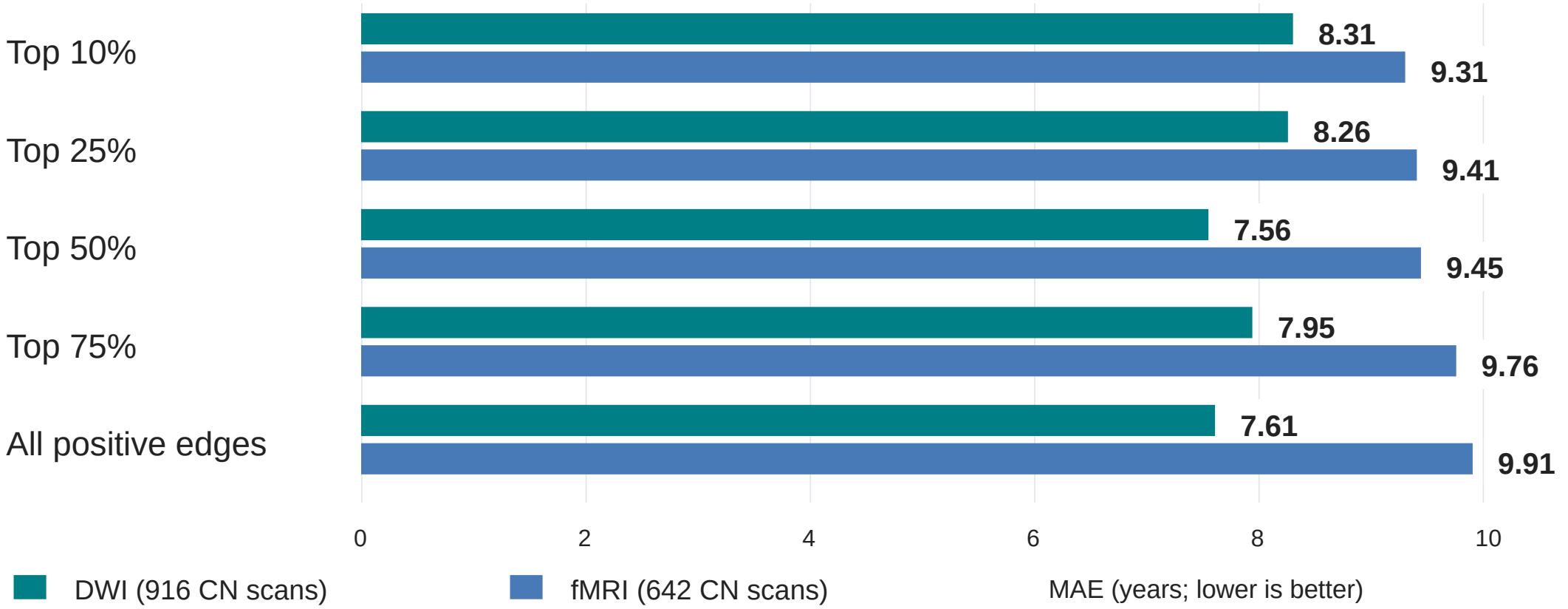
Historical: Ridge leads the full-cohort unimodal analyses

Modality / model	CN scans	MAE (yr)	RMSE (yr)	R^2	Pearson r
DWI Ridge	916	5.98	7.89	0.519	0.724
DWI GNN (top 50%)	916	7.56	9.61	0.287	0.537
fMRI Ridge	642	7.33	9.49	0.406	0.637
fMRI GNN (top 10%)	642	9.31	11.97	0.056	0.305

DWI and fMRI use different populations; compare models within each modality.

DWI_Ridge_CV_metrics.json; fMRI_Ridge_CV_metrics.json; selected GNN threshold summaries

Historical: Graph sparsity has modality-specific effects



Thresholds were selected on the reported OOF comparison; an independent estimate is still needed.

DWI_GNN_threshold_comparison.csv; fMRI_GNN_threshold_comparison.csv

Historical: Paired-sample selection changes the population

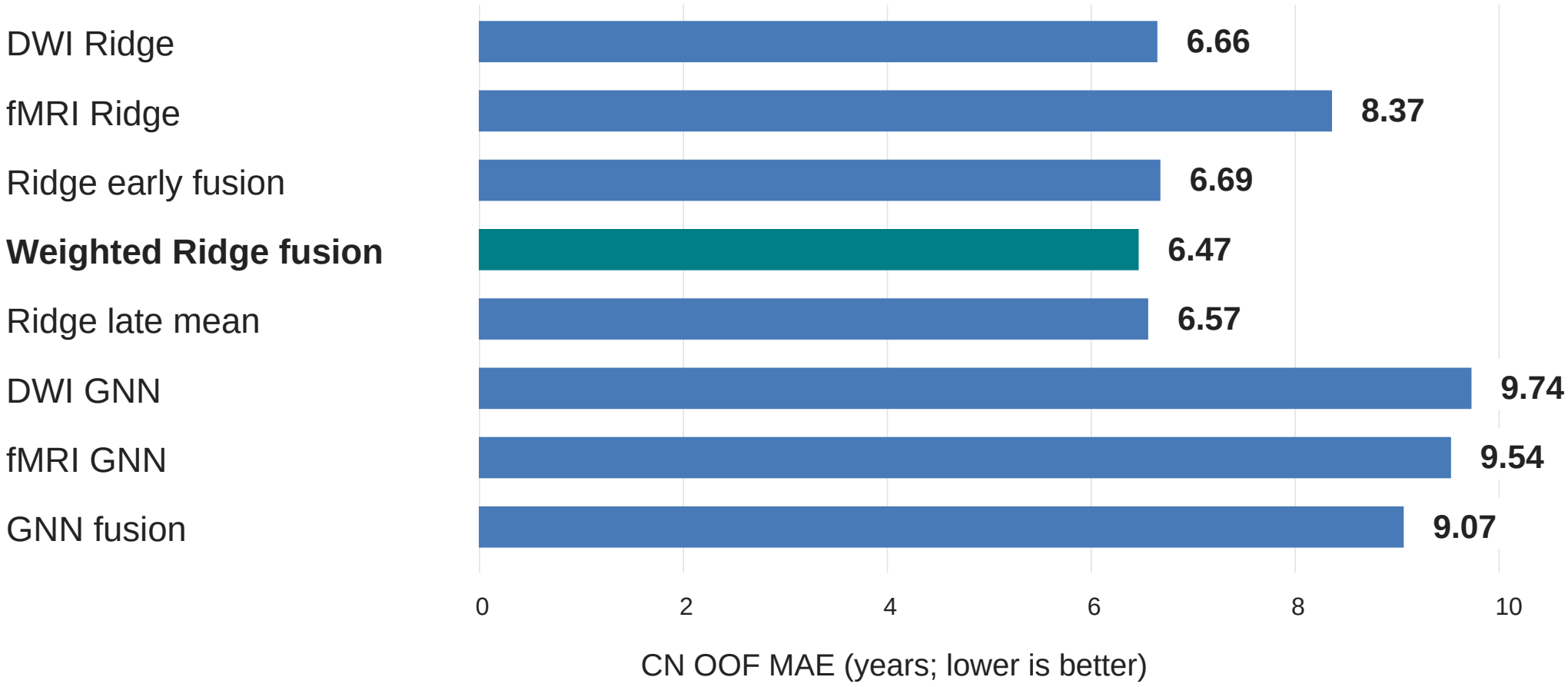
Cohort	Paired scans	Evaluated CN scans	Evaluated participants
ADNI	245	131	74
ADRC	338	243	243
HABS	264	13	13
Total	847	387	330

250 HABS scans have age disagreements between modalities. Training requires ages to agree within 0.01 year; only 13 HABS scans enter this benchmark.

Visit-ID pairing is available; acquisition-date proximity has not been verified.

report_tables/Table1_samples.csv; metadata/fusion_metadata_audit_summary.json

Historical: Weighted Ridge has the lowest paired-sample MAE



Shared benchmark: 387 CN scans / 330 participants / identical outer folds.

report_tables/Table4_models_pooled.csv

Historical: The fusion gain is modest and method dependent

Model	MAE (yr)	RMSE (yr)	R^2	r
DWI Ridge	6.66	8.49	0.620	0.788
Ridge early fusion	6.69	8.62	0.609	0.780
Weighted Ridge fusion	6.47	8.38	0.630	0.794
Ridge late mean	6.57	8.48	0.621	0.799
GNN fusion	9.07	11.44	0.311	0.558

Weighted Ridge improves MAE by 0.184 years (2.8%) vs DWI Ridge. Unweighted concatenation does not improve on DWI alone.

report_tables/Table4_models_pooled.csv; Weighted_Ridge/fusion_summary.json

Historical: Pooled performance masks cohort differences

Cohort	CN scans	DWI Ridge MAE	Weighted Ridge MAE	GNN fusion MAE
ADNI	131	7.15	6.85	8.91
ADRC	243	6.52	6.39	9.04
HABS	13	4.12	4.16	11.24

ADNI R^2 is negative for all paired models despite positive pooled R^2. HABS estimates rely on only 13 scans. These are pooled-training OOF results, not leave-one-cohort-out tests.

report_tables/Table5_models_by_cohort.csv

Historical: Archived attribution: reconstructed fusion GNN

- Integrated gradients (IG) attributes standardized node features and normalized edge attributes on held-out CN scans.
- Five reconstructed outer-fold models explain 387 scans from 330 participants.
- Reference: training-mean node features with observed diagonal retained, and zero edge attributes; graph topology stays fixed.
- Participant summaries average signed attributions across visits before taking absolute values.
- Attributions describe model behavior relative to this reference; they are not causal effects or clinical associations.

Historical results: results/explore/GNN_attribution. Not the output of the current saved-checkpoint workflow.

GNN_interpretability/METHODS.txt; analysis_summary.json

Historical: Attribution quality limits the strength of interpretation

Quality check	Observed result
IG integration steps	64 initially; up to 512
Completeness tolerance	0.05 years
Scans failing completeness	6 / 387 (1.6%)
Maximum absolute IG residual	0.119 years
Maximum difference from original OOF prediction	13.18 years

These attributions explain reconstructed models; exact original-model attribution is not established.

Historical results: results/explore/GNN_attribution. Not the output of the current saved-checkpoint workflow.

GNN_interpretability/analysis_summary.json; reconstruction_check.csv

Historical: fMRI carries larger attribution in this fusion model

Modality	Mean signed IG (yr)	Mean absolute connection IG	Occlusion error increase (yr)
DWI	-6.96	45.17	2.55
fMRI	10.33	128.54	3.36

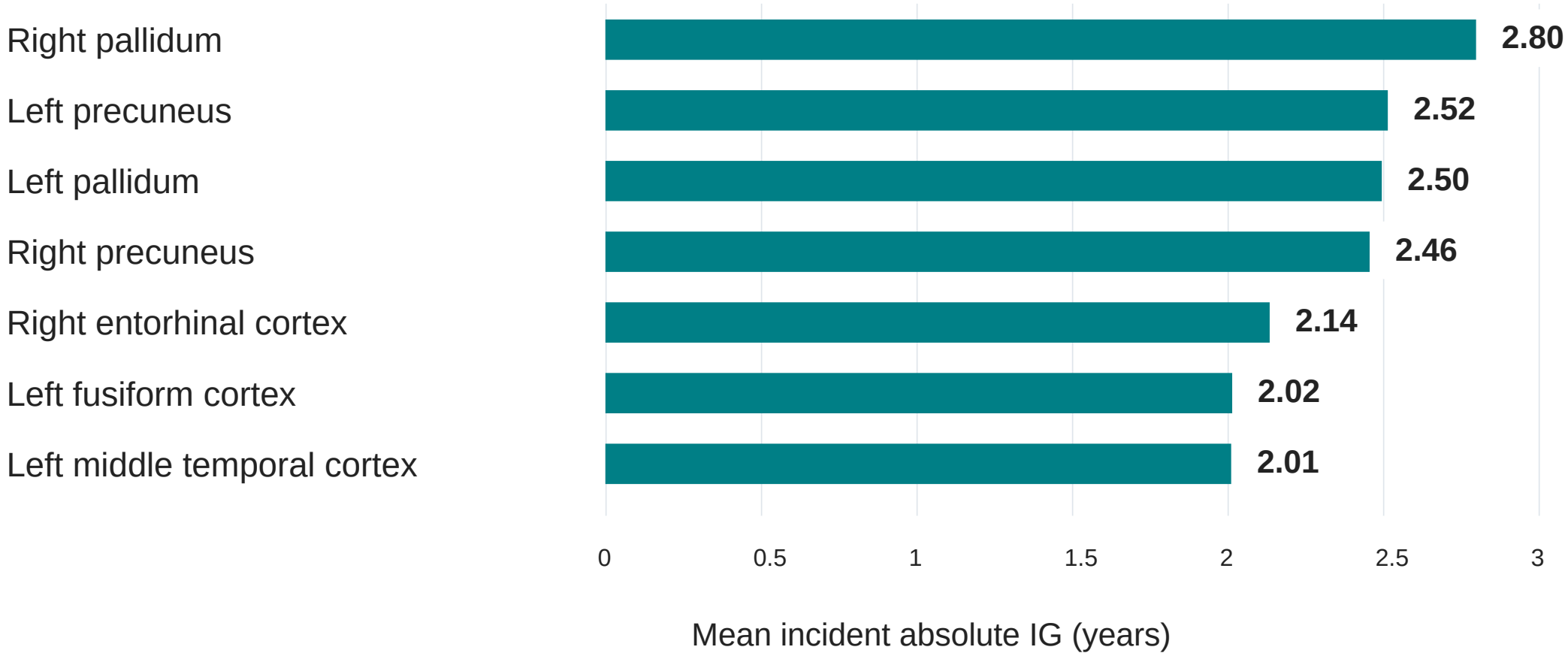
Larger fMRI attribution reflects this model and its reference. It does not imply superior standalone prediction. Absolute connection contributions can accumulate and cancel; they are not predicted ages or modality percentages.

Reference occlusion retains topology and may produce inputs outside the training distribution.

Historical results: results/explore/GNN_attribution. Not the output of the current saved-checkpoint workflow.

GNN_interpretability/modality_summary.csv

Historical: Top ROI attributions emphasize fMRI regions



Top overall rows are fMRI. Region labels follow the existing atlas audit; per-scan order is unverified.

Historical results: results/explore/GNN_attribution. Not the output of the current saved-checkpoint workflow.

GNN_interpretability/roi_importance.csv; roi_labels_with_verification_scope.csv

Historical: Leading connections and stability offer complementary views

Connection (fMRI)	Mean |IG| (yr)	Folds in top 100
lh_precuneus ↔ rh_precuneus	0.415	5/5
lh_entorhinal ↔ rh_entorhinal	0.233	5/5
lh_isthmuscingulate ↔ lh_precuneus	0.195	5/5
lh_inferiorparietal ↔ lh_precuneus	0.183	3/5
lh_insula ↔ rh_insula	0.170	5/5

Ranks describe reconstructed CN models and fixed topology; avoid biomarker claims.

Historical results: results/explore/GNN_attribution. Not the output of the current saved-checkpoint workflow.

GNN_interpretability/connection_importance.csv; connection_fold_stability.csv